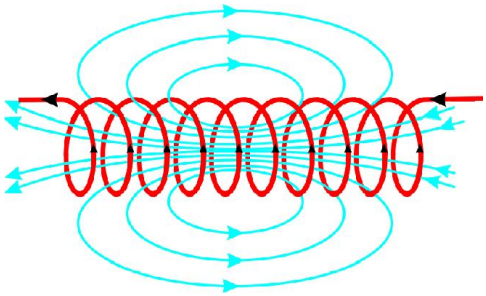
CLASS X · PHYSICAL SCIENCE · CHAPTER 10

Magnetic Effects of Electric Current

Student Workbook – Answer Key

Practise the way the exam asks.

Every topic of the chapter, turned into questions you answer in this book: fill in the blanks, label and draw field lines, apply the hand rules, complete the derivation, solve the numerical, explain the reason. Each unit ends with a self check, and the book ends with a full 40-mark chapter test.



- Marks badges: 1M 2M 4M
- Academic standards: AS1 concept AS3 experiment AS5 diagram AS7 application
- Textbook marks every question from *Improve your learning*

Name _____

Class / Section _____

Roll No. _____

School _____

How to use this workbook

- Read the **Recall in 60 seconds** box first. If a point feels new, go back to the textbook page before you start.
- Answer in the book, in pen. Do the questions **without** looking at the textbook, then check.
- Numericals: *formula → values in SI units (m, m^2, A, T) → answer with unit.*
- Field-line diagrams: draw in pencil, put **arrows on every line**, mark N and S, ⊙ and ⊗.
- For every direction question, write which **rule** you used (thumb rule, right hand rule, Lenz's law).
- Fill in the **Self check** after each unit; copy mistakes into the **Mistake log**.

What the exam asks from this chapter

Marks	Typical questions
1 mark	Units (weber, tesla), one-line laws, energy change in motor/generator, force when parallel to B, MCQs.
2 marks	Direction of current/field/force from a figure, pole of a coil, TV distortion, closed field lines, short numericals on $B = \Phi/A$, $F = ILB$, $\varepsilon = Blv$, $\varepsilon = N\Delta\Phi/\Delta t$.
4 marks	Activities: current produces a field; force on a wire; Faraday's law. Working of motor and generators. Derivation of $\varepsilon = Blv$, $F = ILB$, $r = mv/Bq$. Lenz's law with energy argument. Levitation.
Diagrams	Field lines of bar magnet, straight wire, solenoid; right hand rules; coil in a field; motor; Faraday's activity; AC generator and emf graph.

My progress tracker

Unit	Topic	Score	Confidence	Revised on
1	Oersted, magnetic field and field lines	_____	☐☐☐	_____
2	Magnetic flux and flux density	_____	☐☐☐	_____
3	Fields of a wire, coil and solenoid	_____	☐☐☐	_____
4	Force on a moving charge	_____	☐☐☐	_____
5	Force on a current; electric motor	_____	☐☐☐	_____
6	Induction: Faraday, Lenz, motional emf	_____	☐☐☐	_____
7	Applications, levitation, generators	_____	☐☐☐	_____
–	One-mark rapid fire	_____	☐☐☐	_____
–	Chapter test (40 marks)	_____	☐☐☐	_____

Confidence: tick one box for "shaky", two for "fine", three for "I can teach it".

UNIT 1

Magnetic field and field lines

1M 2M 4M

Recall in 60 seconds

- **Oersted (1820)**: a compass needle under a wire deflects when current flows ⇒ a current produces a magnetic field.
- A **magnetic field** is the region around a magnet where its force acts without contact. It has **strength** and **direction** and is three-dimensional.
- **Field lines** are imaginary; the **tangent** gives the direction; crowded lines = strong field. Outside a magnet they go N → S. Uniform field: parallel, equally spaced lines.

A Fill in the blanks

AS1 1M

1. The scientist who first showed that a current deflects a compass needle was **Oersted**.
2. The direction of the field at a point is given by the **tangent** to the field line at that point.
3. Field lines are **crowded** near the poles, where the field is **strong**.
4. A field whose strength and direction are the same everywhere is called a **uniform** field.
5. Outside a bar magnet field lines go from **N** to **S**.

B True or false

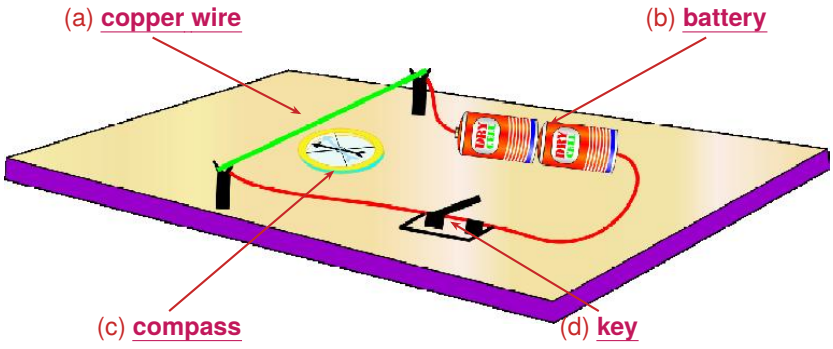
AS1 1M

Write T or F. If false, correct it on the line below.

1. Field lines of a magnet can cross each other. **[F]**
   Correct: They never cross; at a crossing the field would have two directions, which is impossible.
2. The magnetic field of a bar magnet exists only in the plane of the paper. **[F]**
   Correct: It is three-dimensional; it surrounds the magnet in all directions.
3. Far from a magnet a compass needle points almost north–south. **[T]**

C Label the Oersted apparatus

AS5 2M



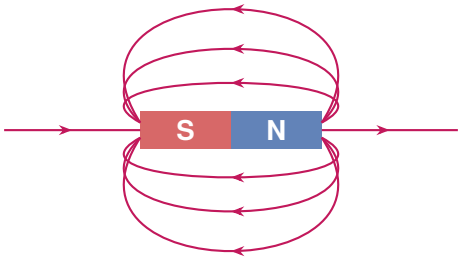
1. What happens to the needle when the key is closed? What does it prove? 2M

   The needle deflects. When the key is opened it returns to the north–south direction. This proves that a current-carrying wire produces a magnetic field around it: electricity and magnetism are related.

D Draw field lines

AS5 2M

Draw at least six field lines of the bar magnet below, with arrows. 2M



E Short answers

Textbook AS1 2M

1. Are magnetic field lines closed? Explain. 2M

Yes. Around a straight wire they are complete circles. For a solenoid (and a bar magnet) the lines outside go from N to S and continue inside from S to N, so every line is a closed loop.

2. How can you find the strength and the direction of a magnetic field using field lines? 2M

Direction: the tangent to the field line at the point (the way a compass needle points). Strength: how close the lines are; crowded lines mean a strong field, widely spaced lines a weak field.

Self check

I can . . .

- ☐ describe Oersted's experiment and its conclusion
- ☐ state the properties of field lines
- ☐ draw the field lines of a bar magnet with arrows

My score

_____ / 19

Redo every question you got wrong on the **Mistake log** page.

UNIT 2

Magnetic flux and flux density

1M 2M

Recall in 60 seconds

- **Magnetic flux** Φ: number of field lines through a plane held perpendicular to the field. Unit **weber (Wb)**.
- **Flux density** $B = \dfrac{\Phi}{A}$ (area perpendicular to the field). Unit **Wb/m^2 = tesla (T)**.
- Plane tilted so its normal makes θ with B: effective area $A\cos\theta$, so $\Phi = BA\cos\theta$. Plane parallel to B: $\Phi = 0$.

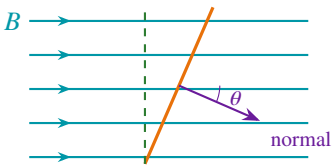
A Fill in the blanks

AS1 1M

1. The SI unit of magnetic flux is the **weber**.
2. The SI unit of magnetic flux density is the **tesla**, which equals **Wb/m^2**.
3. Magnetic flux density is also called magnetic **induction**.
4. The flux through a plane held parallel to the field is **zero**.

B Complete the derivation of $\Phi = BA\cos\theta$

AS1 2M



edge view of a plane of area A

(1) B uses the area **perpendicular** to the field:

$$B = \frac{\Phi}{A_\perp}$$

(2) The effective area perpendicular to the field is $A_\perp = A\cos\theta$.

(3) So $B = \dfrac{\Phi}{A\cos\theta}$ and $\Phi = BA\cos\theta$.

(4) $\theta = 0°$: $\Phi = BA$; $\theta = 90°$: $\Phi = 0$.

C Numericals

AS7 2M

Worked example: 0.02 Wb crosses 0.4 m^2 normally

$B = \dfrac{\Phi}{A} = \dfrac{0.02}{0.4} = 0.05$ T. **Always** convert cm^2 to m^2 (1 cm^2 = 10^{-4} m^2).

1. $B = 0.8$ T passes normally through a loop of area 0.05 m^2. Find the flux. 2M

   $\Phi = BA = 0.8 \times 0.05 = 0.04$ Wb.

2. The same loop is turned so that its normal makes 60° with the field. Find the new flux. 2M

   $\Phi = BA\cos 60° = 0.04 \times \frac{1}{2} = 0.02$ Wb.

3. A flux of 6×10^{-4} Wb passes normally through 20 cm^2. Find B. 2M

   $A = 20 \times 10^{-4} = 2 \times 10^{-3}$ m^2. $B = \dfrac{6 \times 10^{-4}}{2 \times 10^{-3}} = 0.3$ T.

Self check

I can . . .

- ☐ define magnetic flux and flux density with units
- ☐ derive $\Phi = BA\cos\theta$ and use its special cases
- ☐ convert areas to m^2 in numericals

My score

_____ / 15

Redo every question you got wrong on the **Mistake log** page.

UNIT 3

Fields of a wire, a coil and a solenoid

1M 2M 4M

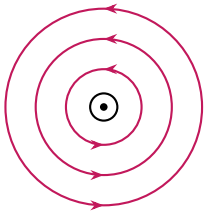
Recall in 60 seconds

- **Straight wire**: field lines are concentric circles. **Right hand thumb rule**: thumb along current, curled fingers give the field. Out of the page ⊙ ⇒ anticlockwise; into the page ⊗ ⇒ clockwise.
- **Circular coil**: field at the centre is perpendicular to the plane of the coil. Curl fingers along the current, thumb gives B. Anticlockwise current (as seen) ⇒ that face is **N**.
- **Solenoid**: behaves like a bar magnet; outside N → S, inside S → N; lines are closed loops.

A Draw the field lines

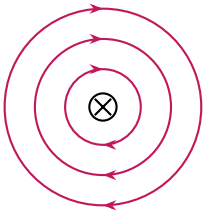
AS5 2M

Draw three field lines with arrows around each wire. 2M



(1) current out of the page

anticlockwise



(2) current into the page

clockwise

B Direction questions from the textbook

Textbook AS1 1M 2M

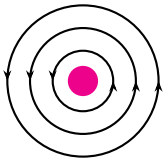


fig-Q1

1. Field lines around a wire are shown in fig-Q1 (anticlockwise). What is the direction of the current? 1M

Out of the page (towards the reader), by the right hand thumb rule.

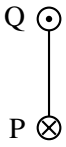


fig-Q3

2. A coil is kept perpendicular to the page. At P the current flows into the page, at Q it comes out (fig-Q3). What is the direction of the field due to the coil? 2M

At the centre, the wire at P (into page, clockwise field) gives a field to the right; the wire at Q (out of page, anticlockwise field) also gives a field to the right. So the field is along the axis of the coil, from left to right, perpendicular to the plane of the coil.

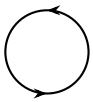


fig-Q4

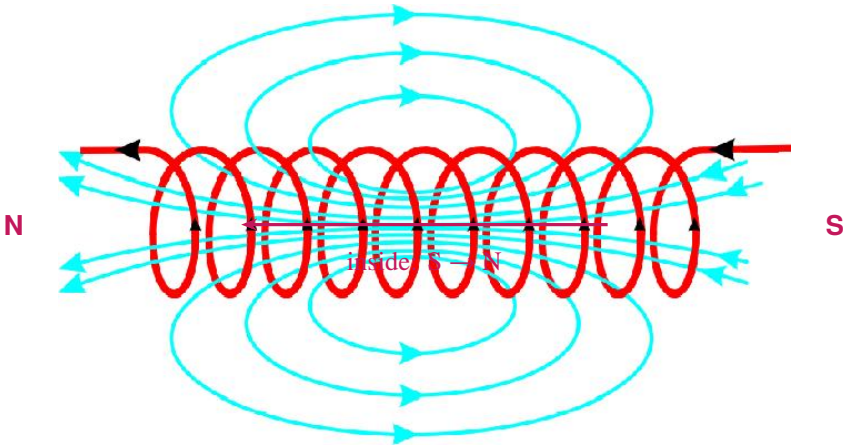
3. The current in a coil flows as in fig-Q4 (anticlockwise as you see it). What pole is formed at this face? 1M

North pole: curling the right-hand fingers anticlockwise, the thumb (field) points towards the viewer.

C Label the solenoid

AS5 2M

In the figure the current enters at the right end and leaves at the left. Mark the N and S ends and the direction of the field inside.



D Experiment

Textbook AS3 4M

Explain with the help of two activities that a current-carrying wire produces a magnetic field. 4M

Activity 1 (Oersted): a wire stretched through the slits of two sticks is connected to a battery through a key; a compass is placed below it. When the key is closed the needle deflects; when it is opened the needle returns to N–S.
Activity 4: a vertical wire passes through a hole in a plank; 6–10 compass needles are placed on a circle around it. When the current flows, the needles set along tangents to the circle; iron filings sprinkled around form concentric circles.
Conclusion: the wire produces a magnetic field whose lines are concentric circles around it; the direction is given by the right hand thumb rule.

Self check

I can . . .

- ☐ draw the field of a straight wire for both current directions
- ☐ find the pole of a coil face from the current direction
- ☐ describe the field of a solenoid and why its lines are closed loops

My score

_____ / 20

Redo every question you got wrong on the **Mistake log** page.

UNIT 4

Force on a moving charge

1M 2M 4M

Recall in 60 seconds

- A magnet distorts the picture on a CRT TV: the field exerts a force on the moving electrons.
- $F = qvB\sin\theta$; $F = qvB$ when $v \perp B$; $F = 0$ when $v \parallel B$ or $v = 0$.
- Direction: right hand rule (forefinger v, middle finger B, thumb F), for a **positive** charge; reverse it for an electron. F is always perpendicular to both v and B.
- A charge moving $\perp B$ goes round a circle: $r = \dfrac{mv}{Bq}$, $T = \dfrac{2\pi m}{Bq}$.

A Fill in the blanks

AS1 1M

1. The force on a charge moving parallel to the field is **zero**.
2. The magnetic force is always perpendicular to both the **velocity** and the **field**.
3. For an electron, the direction of force is **opposite** to that found for a positive charge.
4. The magnetic force on a charge moving $\perp B$ acts as the **centripetal** force.

B Complete the derivation

AS1 4M

A charge q of mass m moves with speed v perpendicular to a uniform field B.

(1) Magnetic force: $F = qvB$ (since $\sin 90° = 1$).

(2) This force is always perpendicular to the velocity, so the path is a **circle**.

(3) Centripetal force = magnetic force: $\dfrac{mv^2}{r} = qvB \quad \Rightarrow \quad r = \dfrac{mv}{Bq}$

(4) Time period $T = \dfrac{2\pi r}{v} = \dfrac{2\pi m}{Bq}$, which does **not** depend on the speed.

C Numericals

AS7 2M

1. A charge of 2×10^{-6} C moves at 3×10^4 m/s perpendicular to $B = 0.5$ T. Find the force. 2M

   $F = qvB = 2 \times 10^{-6} \times 3 \times 10^4 \times 0.5 = 3 \times 10^{-2}$ N.

2. The same charge moves at 30° to the field. Find the force. 2M

   $F = qvB\sin 30° = 0.03 \times \frac{1}{2} = 1.5 \times 10^{-2}$ N.

3. A proton ($m = 1.67 \times 10^{-27}$ kg, $q = 1.6 \times 10^{-19}$ C) moves at 10^6 m/s perpendicular to $B = 1$ T. Find the radius of its path. 2M

   $r = \dfrac{mv}{Bq} = \dfrac{1.67 \times 10^{-27} \times 10^6}{1 \times 1.6 \times 10^{-19}} = 1.04 \times 10^{-2}$ m ≈ 1 cm.

D Short answers

Textbook AS1 2M

1. Why does the picture appear distorted when a bar magnet is brought close to the screen of a television? 2M

The picture on a CRT screen is made by a beam of fast-moving electrons. The magnet's field exerts a force $F = qvB\sin\theta$ on them, which changes their path, so they strike the wrong spots and the picture is distorted.

2. State the right hand rule for the direction of force on a positive charge. **2M**

Stretch the thumb, forefinger and middle finger of the right hand mutually perpendicular. If the forefinger points along the velocity (or current) and the middle finger along the field, the thumb gives the direction of the force.

Self check

I can . . .

- ☐ use $F = qvB\sin\theta$ including the zero-force cases
- ☐ find the direction of force with the right hand rule, and reverse it for electrons
- ☐ derive $r = mv/Bq$ and $T = 2\pi m/Bq$

My score

_____ / 25

Redo every question you got wrong on the **Mistake log** page.

UNIT 5

Force on a current; electric motor

1M 2M 4M

Recall in 60 seconds

- A current is moving charge, so a current-carrying wire in a field feels a force: $F = ILB\sin\theta$; $F = ILB$ when $I \perp B$; $F = 0$ when $I \parallel B$.
- Activity 8: a wire in a horseshoe magnet is deflected; reversing the current or the poles reverses the deflection.
- **Motor**: forces on opposite sides of a coil are equal and opposite (net force 0) but form a **couple** that rotates the coil. **Split rings** and brushes reverse the current every half turn. Electrical → mechanical energy.

A Complete the derivation of $F = ILB$

AS1 4M

(1) Force on one charge moving with drift velocity v perpendicular to B: $F_0 = qvB$.

(2) Total charge in the length L inside the field is Q: $F = QvB$ . . . (1)

(3) Time to cross length L: $t = \dfrac{L}{v}$, so $v = \dfrac{L}{t}$ . . . (2)

(4) Put (2) in (1): $F = \dfrac{Q}{t}LB$ . . . (3)

(5) Since $\dfrac{Q}{t} = I$, $F = ILB$. At an angle θ: $F = ILB\sin\theta$.

B Numericals

Textbook AS7 2M

1. Textbook An 8 N force acts on a 20 cm conductor placed perpendicular to a magnetic field. The current is 40 A. Find the magnetic field induction. 2M

$L = 0.2$ m. $B = \dfrac{F}{IL} = \dfrac{8}{40 \times 0.2} = 1$ T.

2. A wire 0.5 m long carries 4 A at 30° to a field of 0.2 T. Find the force. 2M

$F = ILB\sin 30° = 4 \times 0.5 \times 0.2 \times \frac{1}{2} = 0.2$ N.

C Direction of force

Textbook AS1 2M

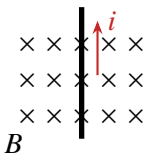


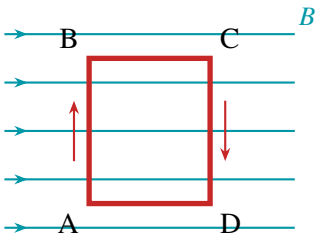
fig-Q5

A straight wire carrying current i upward is kept perpendicular to a field directed into the page; length L is in the field. Find the magnitude and direction of the force. 2M

Magnitude $F = iLB$ (since $\theta = 90°$). Right hand rule: forefinger up (i), middle finger into the page (B) ⇒ thumb points to the **left**. The force acts to the left, in the plane of the page.

D Motor

AS1 AS5 4M



1. The force on AB is directed **into the page** and on CD **out of the page**. 1M

2. The net force on the coil is **zero**, yet the coil **rotates** because the two forces form a **couple**. 1M

3. Why is the current reversed every half turn, and which part does it? 2M

   After half a turn the forces would turn the coil back, so it would only rock to and fro. The split rings (commutator), rotating with the coil, swap brushes every half turn, reversing the current so the rotation continues in the same direction.

E Experiment

Textbook AS3 4M

How do you verify experimentally that a current-carrying conductor experiences a force when kept in a magnetic field? 4M

Materials: wooden plank, two split sticks, copper wire, 9 V battery, switch, horseshoe magnet.
Procedure: pass the wire through the splits of the sticks and connect it to the battery through the switch. Close the switch and bring the horseshoe magnet so the wire lies between its poles.
Observation: the wire is deflected (pushed up or down). Reversing the poles of the magnet, or the direction of the current, reverses the deflection.
Conclusion: a current-carrying conductor in a field experiences a force $F = ILB$; its direction is given by the right hand rule.

Self check

I can . . .

- ☐ derive $F = ILB$ from $F = qvB$
- ☐ find the direction of force on a wire with the right hand rule
- ☐ explain the working of a motor, including split rings

My score

_____ / 27

Redo every question you got wrong on the **Mistake log** page.

UNIT 6

Induction: Faraday, Lenz and motional emf

1M 2M 4M

Recall in 60 seconds

- **Electromagnetic induction**: a current is induced in a coil when the flux through it changes (relative motion of magnet and coil).
- **Faraday's law**: $\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$; for N turns $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t}$.
- **Lenz's law**: the induced current opposes the change in flux (conservation of energy). N pole approaching $\Rightarrow$ near face of coil becomes N.
- **Motional emf**: $\varepsilon = Blv$ for a conductor of length l moving with speed v perpendicular to B.

A Activity 9: what does the galvanometer show?

AS3 1M

Action	Galvanometer
Magnet held at rest inside the coil	no deflection
N pole pushed towards the coil	deflects one way
N pole pulled away from the coil	deflects the opposite way
S pole pushed towards the coil	deflects opposite to N pushed in
Magnet pushed in faster	larger deflection

B Fill in the blanks

AS1 1M

1. The current produced by changing flux is called the induced current.
2. Lenz's law is a consequence of the conservation of energy.
3. When the N pole of a magnet approaches a coil, the face of the coil nearer to it becomes a north pole.
4. $\varepsilon = Blv$ is called the motional emf.

C Complete the derivation of $\varepsilon = Blv$

AS1 4M

A cross wire of length l lies on two parallel conductors in a uniform field B. It is moved a distance s in time Δt and a current I flows.

(1) Force of the field on the cross wire: $F = BIl$ . . . (8)

(2) Work done by us in moving it: $W = Fs = BIls$ . . . (9)

(3) Change in flux as the loop area changes by ls: $\Delta\Phi = Bls$ . . . (10) $\Rightarrow W = I\,\Delta\Phi$

(4) Dividing by Δt: $P = \dfrac{W}{\Delta t} = \dfrac{I\Delta\Phi}{\Delta t}$; comparing with $P = \varepsilon I$ gives $\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$

(5) Also $P = Fv = BIlv$ (since $\dfrac{s}{\Delta t} = v$). So $\varepsilon I = BIlv \Rightarrow \varepsilon = Blv$

D Numericals

Textbook AS7 2M

Worked example (textbook): 400 turns, flux per turn rises by 0.001 Wb in 0.1 s

$\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t} = \dfrac{400 \times 0.001}{0.1} = 4$ V. While the flux stays constant (0.1 s to 0.3 s), $\varepsilon = 0$.

1. Textbook Find the length of a conductor moving at 10 m/s perpendicular to a field of 0.8 T if it induces an emf of 8 V. 2M

$l = \dfrac{\varepsilon}{Bv} = \dfrac{8}{0.8 \times 10} = 1$ m.

2. The flux per turn in a coil of 100 turns falls from 0.05 Wb to 0.01 Wb in 0.2 s. Find the induced emf. 2M

$\Delta\Phi_0 = 0.04$ Wb. $\varepsilon = \dfrac{100 \times 0.04}{0.2} = 20$ V.

3. A 1.5 m rod moves at 2 m/s perpendicular to $B = 0.4$ T. Find the emf. 2M

$\varepsilon = Blv = 0.4 \times 1.5 \times 2 = 1.2$ V.

E Figure questions

Textbook AS1 AS2 2M

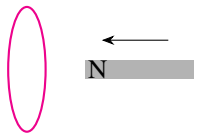


fig-Q2

1. A bar magnet with its N pole facing a coil moves towards it (fig-Q2). What happens to the flux through the coil? Which way does the induced current flow as seen from the magnet? 2M

The flux through the coil increases. By Lenz's law the near face becomes N to repel the magnet, so the induced current is anticlockwise as seen from the magnet.

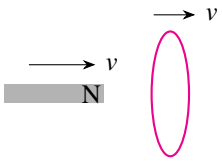


fig-Q6

2. Textbook The coil and the bar magnet both move in the same direction with the same speed (fig-Q6). What happens? 2M

There is no relative motion between the magnet and the coil, so the flux through the coil does not change. No emf is induced and no current flows.

F Long answers

Textbook AS1 AS3 4M

1. Explain Faraday's law of induction with the help of an activity. 4M

Activity: connect a coil to a galvanometer. Push the N pole of a bar magnet into the coil: the needle deflects. Hold it still: no deflection. Pull it out: the needle deflects the other way. With the S pole the deflections reverse. Moving the coil instead of the magnet gives the same result; faster motion gives a larger deflection.
Law: whenever the magnetic flux linked with a closed coil changes, an emf (and current) is induced. The induced emf equals the rate of change of flux: $\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$; for N turns, $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t}$.

2. State Lenz's law and show that it follows from the conservation of energy. 4M

Law: the induced current flows in a direction that opposes the change in flux producing it.
Energy argument: push the N pole towards a coil. If the induced current made the near face S, it would attract the magnet, which would speed up by itself while electrical energy was also produced: energy from nothing, which is impossible. So the near face must be N and repel the magnet; we must do work to push it, and this work becomes electrical energy. Pulling the magnet away, the near face becomes S and attracts it, again opposing the motion.

3. What experiment do you suggest to understand Faraday's law? What instruments are required? Give suggestions for good results and precautions. 4M

Experiment: Activity 9 (magnet moved in and out of a coil connected to a galvanometer).
Instruments: coil of many turns of insulated copper wire, strong bar magnet, sensitive galvanometer, connecting wires.
For good results: use more turns, a stronger magnet and faster motion.
Precautions: tight connections; keep the galvanometer away from the magnet; check the zero before starting; move the magnet along the axis of the coil; do not drop the magnet.

Self check

I can . . .

- ☐ describe Activity 9 and state Faraday's law
- ☐ state Lenz's law and give the energy argument
- ☐ derive $\varepsilon = \Delta\Phi/\Delta t$ and $\varepsilon = Blv$
- ☐ solve induced-emf numericals with N turns

My score

_____ / 44

Redo every question you got wrong on the **Mistake log** page.

UNIT 7

Applications, levitation and generators

1M 2M 4M

Recall in 60 seconds

- Applications of induction: security metal detectors, tape recorder heads, induction stoves, ATM card readers.
- **Levitation**: AC in a coil around an iron core ⇒ changing flux through a metal ring ⇒ induced current opposes it ⇒ repulsion balances the ring's weight. With DC the ring only jumps at switch on/off.
- **AC generator**: coil rotated in a field; full **slip rings**; emf zero when the coil is vertical, maximum when horizontal; reverses every half turn. **DC generator**: half (split) rings keep the output in one direction. Mechanical → electrical energy.

A Match the following

AS1 1M

	Column A		Column B
1.	Induction stove	[c]	a) Induced current in a large coil triggers an alarm
2.	Security check	[a]	b) Magnetised tape moves past a small coil
3.	Tape recorder	[b]	c) Induced current heats a metal pan
4.	Electric motor	[e]	d) Mechanical energy into electrical energy
5.	Generator	[d]	e) Electrical energy into mechanical energy

B Levitation

AS1 AS7 2M 4M

1. Why does the metal ring levitate when AC is used? 4M

The ring is at rest, so the net force on it is zero: an upward force balances its weight. AC changes in size and direction continuously, so the flux through the ring keeps changing and a current is induced in it (Faraday). By Lenz's law this current opposes the change, so the ring's lower face always has the same polarity as the coil's upper face (e.g. coil top S, ring bottom S). The repulsion provides the upward force that holds the ring up.

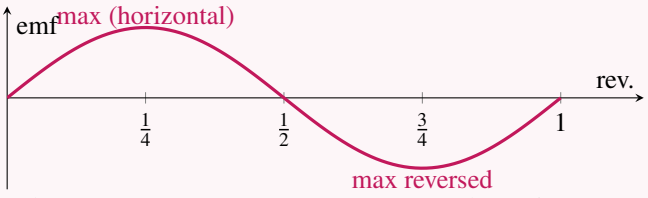
2. What happens if DC is used instead of AC? Why? 2M

The ring jumps up only when the switch is closed (and again when it is opened) and then falls back, because the flux through the ring changes only at those instants. With a steady current there is no change of flux, no induced current and no repulsion.

C AC generator

AS1 AS5 2M 4M

1. Sketch the emf produced by an AC generator over one rotation, starting with the coil vertical. Mark where the emf is zero and maximum. 2M



Zero at 0, $\frac{1}{2}$, 1 rotation (coil vertical); maximum at $\frac{1}{4}$ and $\frac{3}{4}$ (coil horizontal).

2. Why does the emf vary from zero to maximum during a rotation? 2M

The emf depends on how fast the flux through the coil changes. When the coil is vertical (plane ⊥ field) the flux is maximum but changing slowest, so the emf is zero. When it is horizontal (plane ∥ field) the flux is changing fastest, so the emf is maximum.

3. How is an AC generator converted into a DC generator? 2M

Replace the two full slip rings by two half (split) rings. After every half rotation the ends of the coil change brushes, so the current in the external circuit always flows in the same direction (DC).

D Textbook application

Textbook AS7 2M

Give a few applications of Faraday's law of induction in daily life. 2M

Electric generators in power stations; metal detectors at security checks; tape recorder heads; induction stoves; ATM/credit card readers; transformers; bicycle dynamo.

Self check

I can . . .

- ☐ explain four applications of induction
- ☐ explain levitation with AC and why DC fails
- ☐ explain the AC generator, its emf graph and the DC version

My score

_____ / 25

Redo every question you got wrong on the **Mistake log** page.

One-mark rapid fire

Answer in one word or one line. Target **8 minutes** for 25 questions. Time taken: _____ Score: _____ / 25

1. Discovered the magnetic effect of current? **Oersted**
2. Unit of magnetic flux? **weber**
3. Unit of flux density? **tesla**
4. 1 T in N, A, m? **N/(A m)**
5. Φ for a plane $\parallel B$? **zero**
6. Field lines of a straight wire? **circles**
7. Current $\odot$: field lines are? **anticlockwise**
8. Rule for field of a wire? **thumb rule**
9. Anticlockwise current face is? **N pole**
10. Coil that acts like a bar magnet? **solenoid**
11. Force on a charge moving $\parallel B$? **zero**
12. Formula for force on a charge? $qvB\sin\theta$
13. Radius of path of a charge? mv/Bq
14. Force on a current $\perp B$? ILB
15. Motor converts? **elec. → mech.**
16. Reverses current in a motor? **split rings**
17. Faraday's law formula? $\Delta\Phi/\Delta t$
18. Law giving induced current direction? **Lenz's law**
19. Lenz's law is based on? **energy cons.**
20. Motional emf? Blv
21. Emf when coil plane $\perp B$ (generator)? **zero**
22. Rings in an AC generator? **slip rings**
23. Generator converts? **mech. → elec.**
24. Stove using induction? **induction stove**
25. Current reversing periodically? **AC**

Score boosters

Anatomy of a full-marks 4-mark answer

Example question: *Explain the working of an electric motor.*

Part	What to write	Why it matters
1. Principle	A current-carrying coil in a field experiences a force ($F = ILB$).	Scores the concept mark.
2. Diagram	Coil ABCD between N and S, split rings, brushes, battery; arrows for current.	Diagrams carry marks.
3. Forces	AB and CD: equal, opposite forces ⇒ couple ⇒ rotation.	Shows you know why.
4. Commutator	Split rings reverse current every half turn ⇒ continuous rotation.	Key idea examiners look for.
5. Energy	Electrical energy → mechanical energy.	Closes the answer.

Keywords the examiner looks for

Oersted	compass deflects · current produces field
Field lines	tangent gives direction · crowded = strong · closed loops
Straight wire	concentric circles · right hand thumb rule
Force	$qvB\sin\theta$ · $ILB\sin\theta$ · zero when parallel · right hand rule
Motor	couple · split rings · brushes · electrical to mechanical
Faraday	change of flux · rate of change · induced emf
Lenz	opposes the change · conservation of energy
Generator	rotating coil · slip rings · zero/maximum emf · mechanical to electrical

Mistakes that cost marks

- Field lines without arrows.
- θ measured from the plane, not the normal, in $\Phi = BA\cos\theta$.
- Forgetting $F = 0$ when parallel to B.
- Not reversing the force for an electron.
- Using cm or cm^2 without converting.
- Forgetting N in $\varepsilon = N\Delta\Phi_0/\Delta t$.
- Slip rings in a motor (it has split rings).
- No energy argument in Lenz's law.

Formula sheet (cover the right column and test yourself)

Quantity	Formula / unit
Flux density	$B = \dfrac{\Phi}{A}$ tesla = Wb/m^2
Flux	$\Phi = BA\cos\theta$ weber
Force on a charge	$F = qvB\sin\theta$
Circular motion	$r = \dfrac{mv}{Bq}$, $T = \dfrac{2\pi m}{Bq}$
Force on a current	$F = ILB\sin\theta$
Faraday's law	$\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$; $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t}$
Motional emf	$\varepsilon = Blv$

Diagram practice tracker

Draw each diagram three times on separate days. Tick when you can draw it from memory in under 4 minutes.

Diagram	Try 1	Try 2	Try 3
Field lines of a bar magnet	☐	☐	☐
Field of a straight wire (⊙ and ⊗)	☐	☐	☐
Field of a solenoid with N and S	☐	☐	☐
Right hand rule for force	☐	☐	☐
Electric motor with split rings	☐	☐	☐
Faraday's activity (magnet, coil, galvanometer)	☐	☐	☐
AC generator and emf graph	☐	☐	☐

Chapter test

Time: 1 hour 30 minutes **Maximum marks: 40**

Section I: 6 questions × 1 mark. Section II: 5 questions × 2 marks. Section III: 4 questions × 4 marks, each with an internal choice. Section IV: 8 multiple choice questions × 1 mark.

Section I

6 × 1 = 6

1. What is the SI unit of magnetic flux density?

Tesla (Wb/m^2).

2. State Lenz's law.

The induced current opposes the change in flux producing it.

3. What is the force on a current-carrying wire kept parallel to a magnetic field?

Zero.

4. Which device converts mechanical energy into electrical energy?

Generator.

5. Write the formula for motional emf.

$\varepsilon = Blv$.

6. Name the rule used to find the direction of the field around a straight wire.

Right hand thumb rule.

Section II

5 × 2 = 10

7. Why does a TV picture get distorted when a magnet is brought near it?

The picture is made by moving electrons; the magnet's field exerts a force on them and deflects them.

8. A 0.5 m wire carries 2 A perpendicular to a field of 0.4 T. Find the force.

$F = ILB = 2 \times 0.5 \times 0.4 = 0.4$ N.

9. A coil of 50 turns has its flux per turn changed by 0.02 Wb in 0.1 s. Find the induced emf.

$\varepsilon = \frac{50 \times 0.02}{0.1} = 10$ V.

10. Are magnetic field lines closed? Explain.

Yes: circles round a wire; for a solenoid/magnet the lines go N → S outside and S → N inside.

11. A coil and a magnet move together in the same direction at the same speed. Is a current induced? Why?

No: no relative motion, so no change of flux.

Section III

4 × 4 = 16

12. Explain with two activities that a current-carrying wire produces a magnetic field.
OR Describe an activity to show that a current-carrying wire experiences a force in a magnetic field.

See Unit 3, Exercise D or Unit 5, Exercise E.

13. Derive $F = ILB$ for a straight wire in a magnetic field.
OR Derive $r = \frac{mv}{Bq}$ and $T = \frac{2\pi m}{Bq}$ for a charge moving perpendicular to a field.

See Unit 5, Exercise A or Unit 4, Exercise B.

14. Derive $\varepsilon = Blv$ using conservation of energy.
OR State Faraday's law and explain it with an activity.

See Unit 6, Exercise C or Exercise F1.

15. Explain the working of an electric motor with a diagram.
OR Explain the working of an AC generator with the graph of emf for one rotation.

See the anatomy table in Score boosters (motor) or Unit 7, Exercise C.

Section IV

8 × 1 = 8

16. Which converts electrical energy into mechanical energy? [A]

A) motor B) battery
C) generator D) switch

17. The force on a wire perpendicular to a uniform field is [B]

A) 0 B) ILB
C) $2ILB$ D) $ILB/2$

18. One tesla is equal to [B]

A) N/C B) N/(A m)
C) A/m D) N/(A s)

19. Unit of magnetic flux is [D]

A) dyne B) oersted
C) gauss D) weber

20. No force acts on a current-carrying conductor kept [A]

A) parallel to the field B) perpendicular to the field
C) in the field D) away from the field

21. Current flows anticlockwise in a coil as you see it. The face towards you is [B]

A) S pole B) N pole
C) neither D) both

22. In an AC generator the emf is maximum when the plane of the coil is [B]

A) perpendicular to B B) parallel to B
C) at 45° D) at rest

23. The time period of a charge moving in a circle in a uniform field depends on [C]

A) its speed B) the radius
C) m, q and B D) none

My test result

Section	I	II	III	IV	**Total**
Marks	_____/6	_____/10	_____/16	_____/8	_____/40

Mistake log

Write every question you got wrong. Redo it after one week without looking at the answer.

Unit / Q	What I wrote	The correct idea	Redone
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐